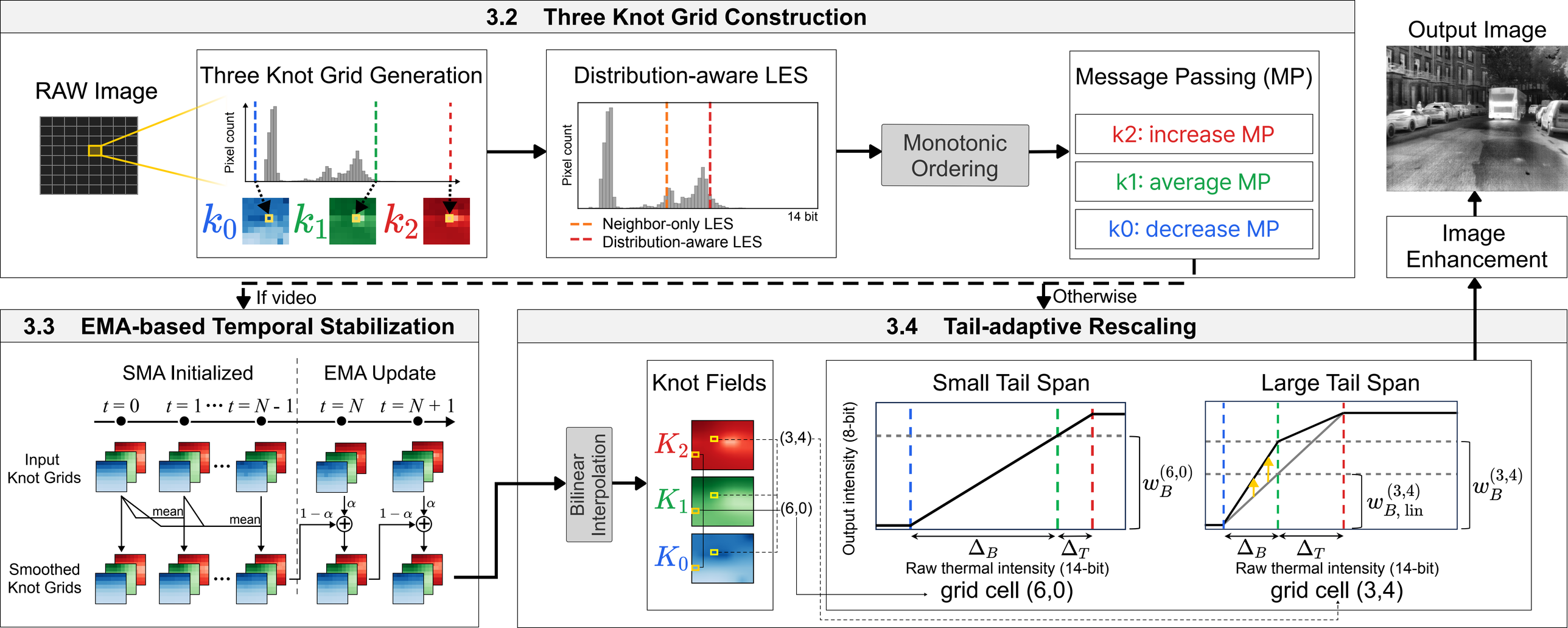

3.2 Three Knot Grid Construction
RAW Image
Three Knot Grid Generation
Pixel count
k_0
k_1
k_2
Distribution-aware LES
Pixel count
14 bit
Neighbor-only LES
Distribution-aware LES
Monotonic Ordering
Message Passing (MP)
k2: increase MP
k1: average MP
k0: decrease MP
Output Image
Image Enhancement
If video
Otherwise
3.3 EMA-based Temporal Stabilization
SMA Initialized
EMA Update
$t = 0$
$t = 1 \cdots t = N - 1$
$t = N$
$t = N + 1$
Input Knot Grids
mean
mean
$1 - \alpha$
α
$1 - \alpha$
α
Smoothed Knot Grids
3.4 Tail-adaptive Rescaling
Bilinear Interpolation
Knot Fields
K_2
K_1
K_0
(3,4)
(6,0)
Small Tail Span
Output intensity (8-bit)
$w_B^{(6,0)}$
Δ_B
Δ_T
Raw thermal intensity (14-bit)
grid cell (6,0)
Large Tail Span
$w_{B,\,\mathrm{lin}}^{(3,4)}$
$w_B^{(3,4)}$
Δ_B
Δ_T
Raw thermal intensity (14-bit)
grid cell (3,4)